

Overview

- Motivation for dimensionality reduction
- Feature selection
 - Wrappers
 - Filters
 - Embedded methods
- Feature transformation
 - Principal Component Analysis (PCA)
 - Autoencoders
 - Clustering

Feature selection

Filters

- We only pick the most informative features for the outcome
- We do not run a machine learning model
- We rank features according to their information and choose a cut-off point
- **Pros**
 - Computationally cheap
- **Cons**
 - No feature interaction is taken into account
 - Machine learning model is not taken into account

k	$J(X_k)$
35	0.846
42	0.811
10	0.810
654	0.611
22	0.443
59	0.388
...	...
212	0.09
39	0.05

Feature selection

Filters - Possible feature evaluation metrics

- **Correlation** of feature x_k with target variable y

$$\rho(x_k, y) = \frac{\text{Cov}(x_k, y)}{\text{Var}(x_k) \text{Var}(y)}$$

Measures **linear** dependencies

- **Mutual information** of feature x_k with target variable y

$$I(x_k, y) = \sum_i \sum_j P(x_k = i, y = j) \frac{P(x_k = i, y = j)}{P(x_k = i)P(y = j)}$$

where P is the probability estimate from the data

Assumes **known probability distribution** of the data.

Feature selection

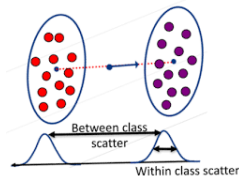
Filters - Possible feature evaluation metrics

- **Fisher's criterion** of feature d of sample $\mathbf{x}_n$ from class k

$$F(d) = \frac{\sum_k \sum_{\mathbf{x}_n \in C_k} (x_{nd} - \mu_{kd})^2}{\sum_k (\mu_d - \mu_{kd})^2}$$

where μ_{kd} is the mean of feature d from class k , and μ_d is the mean of feature d from all samples

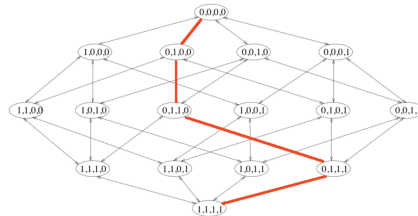
Measures within-class scatter in relation to between-class scatter



Feature selection

Filters

- A lot less expensive



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Feature selection

Embedded methods

- The classifier performs feature selection as part of the learning procedure
- Regularization is a great example

$$J(\mathbf{w}) = EC(\mathbf{w}) + \lambda \sum_{d=1}^D w_d^2 = EC(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

- **Pros**
 - Feature selection is part of learning the procedure
- **Cons**
 - Computationally demanding